

YEAR 11 PHYSICS ATAR
Investigation 11.1 - Investigating V and I
Marking Rubric

NAME: ALEX SKILOVITIS

		Marks Allocated	Mark
Aim	Clear aim given. <i>X Not written. No mention of finding resistance relationship.</i>	1	0
Method	Refers to laboratory manual for method.	2	2
	Drawing of circuit. ✓		
Results	Headings and units. <i>Units in data - not headings.</i>	3	2
	Appropriate columns (V, I) ✓		
	Correct significant figures. $\frac{1}{2}$ <i>Small error.</i>		
Graph	3 graphs drawn ✓	5	5
	Correct labels and units - both axes. ✓✓		
	Plotting. ✓		
	Line of best fit ✓		
Gradient	Uses 2 points of line of best fit. ✓	5	5
	Correct calculation. ✓		
	Correct significant figures. ✓✓		
	Correct units. ✓		
	3 gradients correct ✓		
Discussion	Q3 - were any resistors 'ohmic'? ✓	3	2½
	Q4 - why are the graphs different? ✓		
	Q5 - types of ohmic materials <i>what about the globe?</i>		
Conclusion	Statement addresses the aim. <i>X No conclusion</i>	1	0
		TOTAL	$\frac{16\frac{1}{2}}{20}$

A good report Alex, but there are some errors that need to be eradicated, such as putting units in the data rather than the heading in a table. Your values for the resistors are significantly inaccurate - should be 33 Ω and 15 Ω .

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NAME: ALEX FALCK

		Marks Allocated	Mark
Aim	Clear aim given. <i>No mention of finding unknown resistance.</i>	1	$\frac{1}{2}$
Method	Refers to laboratory manual for method. ✓	2	2
	Drawing of circuit. <i>Not a drawing.</i>		
Results	Headings and units. <i>Some mistakes. X</i>	3	2
	Appropriate columns (V, I) ✓		
	Correct significant figures. ✓		
Graph	3 graphs drawn ✓	5	5
	Correct labels and units - both axes. ✓✓		
	Plotting. ✓		
	Line of best fit ✓		
Gradient	Uses 2 points of line of best fit. ✓	5	$4\frac{1}{2}$
	Correct calculation. ✓		
	Correct significant figures. $\frac{1}{2}$ gave incorrect.		
	Correct units. ✓		
	3 gradients correct ✓		
Discussion	Q3 - were any resistors 'ohmic'? ✓	3	3
	Q4 - why are the graphs different? ✓		
	Q5 - types of ohmic materials ✓		
Conclusion	Statement addresses the aim. $\frac{1}{2}$	1	$\frac{1}{2}$
		TOTAL	$\frac{17\frac{1}{2}}{20}$

A very good report Alex - just a couple of small things that need to be rectified in future practicals. Your values for the resistors are very accurate - well done.

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NAME: ALEX RANKIN

		Marks Allocated	Mark
Aim	Clear aim given. $\frac{1}{2}$ <i>no mention of determining the unknown resistance!</i>	1	$\frac{1}{2}$
Method	Refers to laboratory manual for method. ✓	2	2
	Drawing of circuit. <i>Not a drawing.</i>		
Results	Headings and units. <i>Small mistake.</i>	3	$2\frac{1}{2}$
	Appropriate columns (V, I) ✓		
	Correct significant figures. ✓		
Graph	3 graphs drawn ✓	5	5
	Correct labels and units - both axes. ✓✓		
	Plotting. ✓		
	Line of best fit ✓		
Gradient	Uses 2 points of line of best fit. ✓	5	$4\frac{1}{2}$
	Correct calculation. ✓		
	Correct significant figures. $\frac{1}{2}$ <i>Incorrect on resistor 1!</i>		
	Correct units. ✓		
	3 gradients correct ✓		
Discussion	Q3 - were any resistors 'ohmic'? ✓	3	$2\frac{1}{2}$
	Q4 - why are the graphs different? <i>Heating effect?</i> $\frac{1}{2}$		
	Q5 - types of ohmic materials ✓		
Conclusion	Statement addresses the aim. ✓	1	1
		TOTAL	$\frac{18}{20}$

A very good report Alex. Your values for the resistors are very accurate - well done.

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NAME: COURTNEY GRAY

		Marks Allocated	Mark
Aim	Clear aim given. ✓	1	1
Method	Refers to laboratory manual for method. ✓	2	2
	Drawing of circuit. ✓		
Results	Headings and units. ✓	3	2½
	Appropriate columns (V, I) ✓		
	Correct significant figures. ½ ✓		
Graph	3 graphs drawn ✓	5	5
	Correct labels and units - both axes. ✓✓		
	Plotting. ✓		
	Line of best fit ✓		
Gradient	Uses 2 points of line of best fit. ✓	5	5
	Correct calculation. ✓		
	Correct significant figures. ✓		
	Correct units. ✓		
	3 gradients correct ✓		
Discussion	Q3 - were any resistors 'ohmic'? light globe isn't.	3	1½
	Q4 - why are the graphs different? ✓		
	Q5 - types of ohmic materials ✗		
Conclusion	Statement addresses the aim. ✓	1	1
		TOTAL	<u>18</u> 20

A very good report Courtney. Your values for the resistors are quite accurate - well done!

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NAME: MYA UBALDE

		Marks Allocated	Mark
Aim	Clear aim given. ✓	1	1
Method	Refers to laboratory manual for method. ✓	2	2
	Drawing of circuit. ✓		
Results	Headings and units. ✓	3	3
	Appropriate columns (V, I) ✓		
	Correct significant figures. ✓		
Graph	3 graphs drawn ✓	5	5
	Correct labels and units - both axes. ✓✓		
	Plotting. ✓		
	Line of best fit ✓		
Gradient	Uses 2 points of line of best fit. ✓	5	5
	Correct calculation. ✓		
	Correct significant figures. ✓		
	Correct units. ✓		
	3 gradients correct ✓		
Discussion	Q3 - were any resistors 'ohmic'? ✓	3	2
	Q4 - why are the graphs different? ✓ <i>heating effect?</i>		
	Q5 - types of ohmic materials ✗		
Conclusion	Statement addresses the aim.	1	1
TOTAL			<u>19</u> 20

An excellent report Mya. Your experimental values for the resistors are very accurate - well done.

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NAME: MARIA BRAVO

		Marks Allocated	Mark
Aim	Clear aim given. <i>Second part incorrect.</i>	1	$\frac{1}{2}$
Method	Refers to laboratory manual for method. ✓	2	2
	Drawing of circuit. ✓		
Results	Headings and units. ✓	3	3
	Appropriate columns (V, I) ✓		
	Correct significant figures. ✓		
Graph	3 graphs drawn ✓	5	$4\frac{1}{2}$
	Correct labels and units - both axes. ✓✓		
	Plotting. ✓		
	Line of best fit <i>light grey should have been better.</i>		
Gradient	Uses 2 points of line of best fit. ✓	5	$4\frac{1}{2}$
	Correct calculation. ✓✓✓		
	Correct significant figures. X ✓✓		
	Correct units. ✓		
	3 gradients correct ✓		
Discussion	Q3 - were any resistors 'ohmic'? ✓	3	2
	Q4 - why are the graphs different? X		
	Q5 - types of ohmic materials ✓		
Conclusion	Statement addresses the aim. ✓	1	1
		TOTAL	$17\frac{1}{2}$ 20

A very good report Maria. Your values for the resistors are very accurate - well done.

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NAME: JESSE HUGHES

		Marks Allocated	Mark
Aim	Clear aim given. <i>Don't make sense. No mention of determining unknown resistance</i>	1	$\frac{1}{2}$
Method	Refers to laboratory manual for method. ✓	2	2
	Drawing of circuit. ✓		
Results	Headings and units. ✓	3	3
	Appropriate columns (V, I) ✓		
	Correct significant figures. ✓		
Graph	3 graphs drawn ✓	5	5
	Correct labels and units - both axes. ✓ ✓		
	Plotting. ✓		
	Line of best fit ✓		
Gradient	Uses 2 points of line of best fit. ✓	5	$4\frac{1}{2}$
	Correct calculation. ✓		
	Correct significant figures. X <i>factor 1 incorrect.</i> ✓ ✓		
	Correct units. ✓		
	3 gradients correct ✓		
Discussion	Q3 - were any resistors 'ohmic'? ✓	3	2
	Q4 - why are the graphs different? X <i>Not answered correctly.</i>		
	Q5 - types of ohmic materials ✓		
Conclusion	Statement addresses the aim. X	1	0
		TOTAL	$\frac{17}{20}$

*Your values are very accurate Jesse - well done.
 No conclusion given, but generally done very well.*